

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the —README.md— for this assignment includes instructions to regenerate this handout with your typeset L^AT_EX solutions.

1.b Behavior Cloning Results

Both environments were tested with identical hyperparameters for fair comparison:

- Network: 2 hidden layers, 64 units each
- Training steps: 1000 gradient updates
- Learning rate: 5e-3
- Eval batch size: 5000 (for multiple rollouts)

Environment	Expert Return	BC Mean	BC Std	% of Expert
Ant-v4 (working)	4713.65	1551.48	993.03	32.9%
Walker2d-v4 (not working)	5566.85	295.01	341.21	5.3%

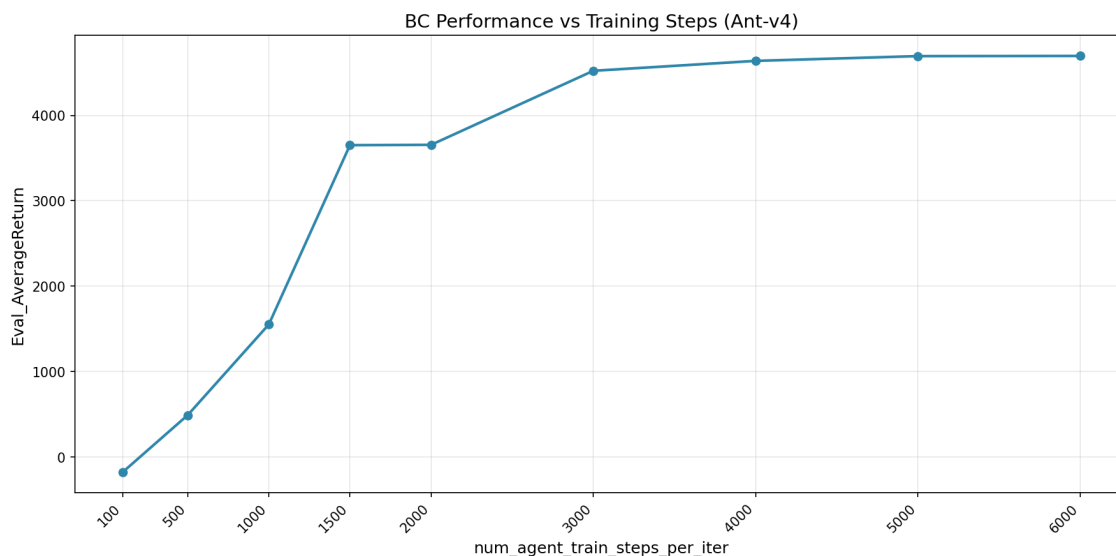
Analysis: BC achieves >30% on Ant-v4 but fails on Walker2d-v4 (only 5.3%). Walker2d requires precise balance and timing that BC struggles to learn from static demonstrations. The high standard deviation in Walker2d (341.21) indicates inconsistent behavior, with episodes often terminating early (avg episode length: 126 vs 1000 for expert).

1.c

Hyperparameter: `num_agent_train_steps_per_iter`

Rationale: This hyperparameter controls how many gradient updates the policy network receives during training. It directly affects the underfitting/overfitting trade-off: too few steps means the policy hasn't learned the expert behavior, while too many steps could lead to overfitting on the limited expert data.

Experiment: Tested on Ant-v4 with values: 100, 500, 1000, 1500, 2000, 3000, 4000, 5000, 6000.



Results: Performance increases rapidly from 100 to 3000 steps, then plateaus near expert level (~ 4700). At 100 steps, the policy is severely underfit (negative return). No overfitting was observed up to 6000 steps, suggesting the expert dataset is sufficient for this task.

2.b